

Placement Portal Application - Project Report

Student Details

Name: Parth Sharma

Roll Number: [2xfxxx44x8](#)

Email: 2xfxxx44x8@ds.study.iitm.ac.in

About: Hi, I'm Parth, a Data Science student at IIT Madras. I have a strong interest in web development and love working on full-stack projects, especially using Python

Project Details

Project Understanding:

- I developed a web-based Placement Portal to simplify and organize the campus recruitment process for students, companies, and the institute.
- Implemented role-based access control (Admin, Student, Company) with distinct functionalities for each role
- Implemented REST API endpoints for students, drives and applications.
- Designed a complete workflow for:
 - Job posting by recruiters
 - Student applications
 - Application tracking and status updates
 - Resume management

AI/LLM Usage

Tools Used: Google Gemini

- Take help in debugging errors and improving code correctness (<1%)
 - Take help in styling UI using CSS and Bootstrap (8%)
 - Take help in implementing charts using Chart.js (3%)
 - Take help in understanding concepts and improving overall code structure
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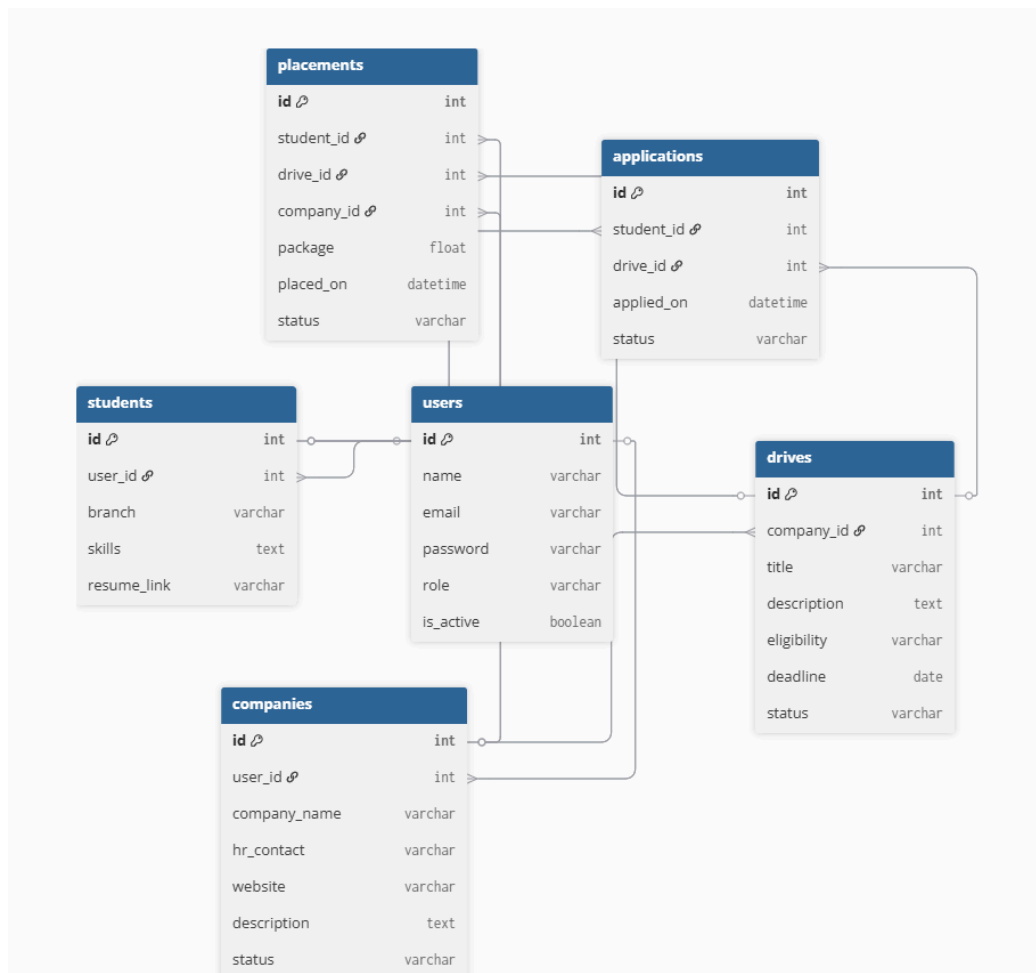
Technologies Used

- **Backend:** Flask, Flask-SQLAlchemy, Flask-Login
- **Frontend:** Html, CSS, Jinja2, Bootstrap
- **Database:** SQLite

DB Schema Design

Tables and Structure:

- **USER**
 - **Relationships:**
 - One-to-One with Student
 - One-to-One with Company
- **STUDENT**
 - **Relationships:**
 - One-to-One with User
 - One-to-Many with Application
 - One-to-Many with Placement
- **COMPANY**
 - **Relationships:**
 - One-to-One with User
 - One-to-Many with Drive
 - One-to-Many with Placement
- **DRIVE**
 - **Relationships:**
 - Many-to-One with Company
 - One-to-Many with Application
 - One-to-Many with Placement
- **APPLICATION**
 - **Relationships:**
 - Many-to-One with Student
 - Many-to-One with Drive
- **PLACEMENT**
 - **Relationships:**
 - Many-to-One with Student
 - Many-to-One with Drive
 - Many-to-One with Company



Route Design

Authentication:

- GET /login – open login page
- POST /login – check login details
- GET /register/student – open student register page
- POST /register/student – create student account
- GET /register/company – open company register page
- POST /register/company – create company account
- GET /logout – log out user

Admin:

- GET /admin/dashboard – open admin dashboard
- GET /admin/students – view all students
- GET /admin/companies – view all companies
- GET /admin/toggle_user/<id>/<source> – enable/disable user
- GET /admin/company/approve/<id> – approve company

- GET /admin/company/reject/<id> – reject company
- GET /admin/drives – view all drives
- GET /admin/drive/approve/<id> – approve drive
- GET /admin/drive/close/<id> – close drive
- GET /admin/applications – view all applications
- GET /admin/placements – view all placements
- GET /admin/search – open search page
- POST /admin/search – search students/companies
- POST /admin/company/blacklist/<id> – delete company
- POST /admin/student/blacklist/<id> – delete student

Student:

- GET /student/dashboard – open student dashboard
- GET /student/drives – view available drives
- GET /student/apply/<drive_id> – apply for drive
- GET /student/applications – view applied drives
- GET /student/profile – view profile
- POST /student/profile – update profile

Company:

- GET /company/dashboard – open company dashboard
- GET /company/create_drive – open create drive page
- POST /company/create_drive – create new drive
- GET /company/drives – view created drives
- GET /company/drive/edit/<id> – open edit drive page
- POST /company/drive/edit/<id> – update drive
- GET /company/drive/close/<id> – close drive
- GET /company/drive/delete/<id> – delete drive
- GET /company/applicants – view applicants
- GET /company/update_application/<id>/<status> – update application status
- POST /company/mark_selected – mark student placed
- GET /company/placements – view placements

API Endpoints:

- GET /api/drives – get all approved drives (JSON)
- GET /api/students – get all students (JSON)
- GET /api/drives/<drive_id>/applications – get applications for a drive (JSON)
- GET /api/students/<student_id> – get student profile (JSON)

Architecture and Features

Architecture Pattern:

- **app.py** – main Flask application entry point
- **models.py** – database models using SQLAlchemy
- **/routes** – Flask Blueprints for admin, student, and company routes
- **/templates** – Jinja2 HTML templates for UI
- **/static** – CSS and static files

Features Implemented:

- **User Authentication & Authorization:**
 - o Secure login and registration system
 - o Password hashing using Werkzeug
 - o Role-based access control (Admin, Student, Company)
 - o Session management using Flask-Login
- **Admin Features:**
 - o View and manage all students and companies
 - o Approve or reject company registrations
 - o Approve or close placement drives
 - o View all applications and placements
 - o Search students and companies
 - o Blacklist/remove students and companies
 - o View dashboard statistics
- **Student Features:**
 - o Register and log in to the system
 - o View available placement drives
 - o Apply for placement drives
 - o View application status
 - o Upload resume and update profile
 - o View applied drives and history
- **Company Features:**
 - o Register and create company profile
 - o Log in after admin approval
 - o Create placement drives
 - o Edit, delete, or close drives
 - o View applicants for drives
 - o Update application status (Shortlisted/Selected/Rejected)
 - o Mark students as placed
 - o View placement records

Project Demo Video: [Mad 1 Project_Demo.mp4](#)